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Digital Style Design of Nanjing Brocade Based on Deep Learning

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***Abstract*—Nanjing Brocade is a traditional Chinese silk fabric with precious materials and exquisite patterns. As the world's intangible cultural heritage, it is essential to inherit and innovate Nanjing Brocade by digital methods. Digital protection of Nanjing Brocade includes cultural connotation, pattern and technology et al. In view of pattern digitization, this paper proposes a deep learning model which is utilized to transfer Yunjin style to modern patterns. The generated images have been applied to the textile productions, and the results show that the proposed model can be adopted by the Yunjin pattern design.**

Keywords—component: nanjing brocade; digital protection; deep learning; style transfer

I. NANJING BROCADE AND DIGITAL PROTECTION

Among the ancient silk fabrics, "brocade" represents the highest technical level of the fabric, while Nanjing Brocade (also named Yunjin) is the most famous in ancient China by absorbing the advantages of brocade weaving techniques of all dynasties. Nanjing Brocade is different from other brocades, in which the specific aspects are as follows:

A. Material

Silk is mainly uesed as the raw material in Common brocade. Besides silk, Yunjin also has the application of other special materials such as gold thread, silver thread, and peacock feather. The application of these materials makes the silk fabrics more gorgeous, unique and magnificent.

B. Pattern

Symbolism and homophony are often used to create the pattern of Yunjin, and the generated pattern represents the auspicious implication. So, Chinese summrize the Yunjin pattern with the following sentence -- "The pattern must be intentional, meaning must be auspicious." In terms of color, the dominant hue of Yunjin is bright.This is mainly relevant to the application of gold and silver thread.The two kinds of luster color can make the fabric more splendid and brilliant.

C. Technology

The core technology of Nanjin Brocade is "warp passing and weft breaking", that is, the warp lines between two patterns are interlinked, and the weft lines are inserted in different lines according to the color. This method can not only meet the requirements of abundant colors, but also reduce the thickness of the fabric. It takes two workers to weave on a large, complex loom. The weavers have to memorize dozens of tips and can only weave 5 to 6 centimeters of brocade a day. Thus, Nanjing Brocade has been known as "Inch Brocade Inch Gold" since ancient times.

Because of its noble characteristics, Yunjin was highly sought after by feudal rulers and reached its peak in the Ming and Qing Dynasties. During the period of the Republic of China, a large number of imperialist woolen cloth was dumped in China, as well as the promotion of mechanized production, which led to the gradual decline of Yunjin. Before and after liberation, Yunjin industry almost all stopped production. After 1949, the government has invested sizable capital to restore and protect Yunjin, but it is still facing the situation of no successor, only 50 people really master the technology of Yunjin. In order to save Nanjing Brocade from extinction, Jiangsu Provincial Government approved the establishment of "Nanjing Brocade Research Institute", which is the only professional institution in China that integrates research, production, display and sales of Yunjin. The related researchers collected and sorted out the patterns and drawings of Yunjin and trained apprentices, which gradually restored the weaving technology of it. Until 2005, the State Council issued a document to strengthen the protection of China's intangible cultural heritage. In September 2009, Nanjing Yunjin wood manual weaving technique was successfully added into the United Nations Intangible Heritage Representative List.

In the digital era, the emergence and wide application of digital technology have changed the way people protect intangible cultural heritage. For the digital protection of Yunjin, Lu [1] designed a mobile application on the iOS

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platform aimed at popularizing the basic knowledge and technological process of Yunjin, and used virtual reality technology to reproduce the weaving of it, creating an immersive interactive experience. Cao and Zhang [2] briefly discussed the specific application methods of digital protection of Yunjin from aspects of network publicity, digital museum and e-commerce platform.

These digital practices of intangible cultural heritage protection tend to regard digital technology as a form of external intervention, emphasizing the use of digital technology to build archives, websites and textbooks for intangible cultural heritage, but pay little attention to how to internalize digital technology. Later researchers have supplemented this aspect. For example, Yi [3] started with the lack of innovation of Nanjing Brocade pattern, and realized the innovation of Nanjing Brocade pattern style by using computer-aided design software. Qiu [4] et al. proposed to cut the existing Yunjin pattern, extract the pattern, and establish a pattern database, so as to reconstruct and innovate the pattern in the later stage.

Although this kind of methods realize the digital innovation of intangible cultural heritage, it is still in a relatively low stage of pattern design and is not conducive to the inheritance of the overall style of Yunjin. In recent years, the rapid development of deep learning provides new methods for image processing and design, as well as new ideas for the design of Yunjin patterns.

II. DEEP LEARNING STYLE TRANSFER ALGORITHM

Convolutional neural network (CNN) model is one of the typical deep learning models, which is widely used in style transfer. The model generally has three operations: convolution, pooling and activation. Various feature extraction filter matrices which also called as convolution kernels are usually set as a size of 3×3 or 5×5. Then the convolutional kernels are used to 'slide' on the original image matrix (the image is actually a matrix composed of pixel values), which is performed to obtain a feature map. The number of convolution kernels determines the count of feature maps.

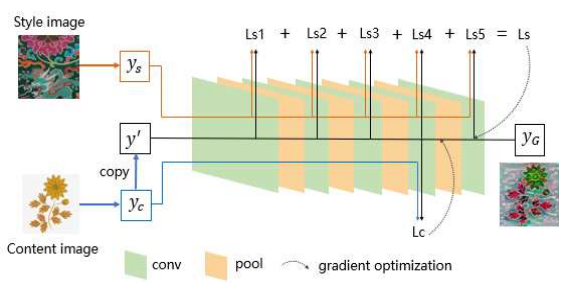

Figure 1. Style transfer algorithm model

In 2015, Gatys [5] et al. proposed an algorithm of style transfer based on the convolutional neural network model: Input two images, one for content, denoted as *C*, and the other for style, denoted as *S*. Firstly, duplicate the content image as the base image. Then, the features of content image and style image are extracted respectively by convolutional network. Finally, a generated image *Ig* is obtained, which contains the content of image *C* and the style of image *S*. The way to get the generated image is optimizing both Content Loss and Style Loss by gradient discent. The specific model is shown in Figure 1. The methods of calculating the two losses are as follows:

- Content Loss (*Lc*): The feature maps of *Ic* and *Ig* on the fourth layer are selected, and the square error between the two feature maps is the Content Loss.
- Style Loss (*Ls*): The correlation between the features of the same layer is used to describe the texture. This correlation can be calculated by the Gram Matrix, that is, the product of the output feature map matrix of a certain layer and its transpose matrix. The Gram matrices of *Is* and *Ig* are calculated respectively, and their square error is denoted as Style Loss.

The total loss (*Lt*) is the linear combination of *Lc* and *Ls*, as shown in (1). Where α and β are the weighting factors for content and style reconstruction respectively.

$$Lt = \alpha \cdot Lc + \beta \cdot Ls \tag{1}$$

III. THE STYLE TRANSFER OF YUNJIN

As a traditional silk fabric, many ancient patterns are used in Nanjing Brocade, such as dragons, phoenixes and so on. On the whole, there is still a certain gap with the modern aesthetic of patterns. To keep the style of Nanjing Brocade, the algorithm is applyed to transfer the style of Nanjing Brocade to modern patterns, which combines the inheritance of Nanjing Brocade with modern aesthetics.

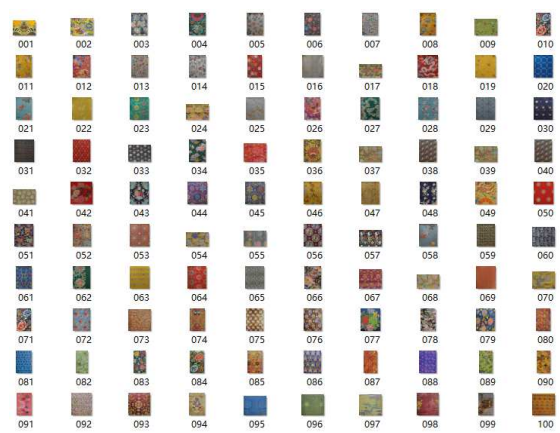

Figure 2. The dataset of Yunjin

Aiming at the digitization of Yunjin patterns, more than 500 pictures are obtained by on-the-spot photography of Nanjing Institute of Yunjin and scanning of related books. Considering the similarity of color, 100 representative pictures were selected as the dataset of Yunjin, as shown in Figure 2. According to the characteristics of the pictures, the modern patterns are divided into two categories, color block and line draft, with 20 pictures in each category.

In this paper, the transfer model is trained on above dataset. Our implementation uses Pytorch and CUDA. During training, to extract image information on comparable scales, we resize the style image to the same size as the content image. We train our network with a batch size of 1 and use L-BFGS to optimize numerical value, which works best for image synthesis. The optimization are terminated at 800 iterations, where results have no visible changes with further iterations. The loss values of the whole procedure are

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plotted as the loss curve shown in Figure 3. A generated image will be output each 100 iterations. Generated images of 100, 200, 300 and 500 times of training were selected as references, as shown in Figure 4.

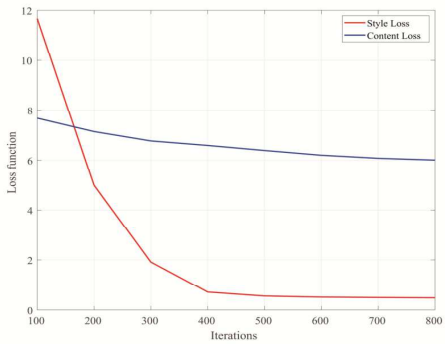

Figure 3. The loss curve

(a) 100 times (b) 200 times (c) 300 times (d) 500 times

Figure 4. Generated graphs of different training times

Comparing these generated images, we can find that with the increase of iterations, the color style becomes more and more deep. Designers can choose images from different levels according to their design style.

A

B

C

(a) Style image (b) Content image (c) Generated image

Figure 5. The effect of style transfer

The transfer model after training is applied, and the effect of style transfer is shown in Figure 5. We set three group: the content of group A is the color block, group B is the line draft, group C is the supplementary experimental group. By comparing group A and group B, it can be found that the color blocks of the generated images in group A are relatively clear, while the colors of the generated images in group B are mixed. This result puts forward a conjecture: although the process of style transfer is not controllable, the transfer will be affected by the color pixel, resulting in the effect of corresponding transfer. Experimental group C added color blocks to the line draft, and the transfer effect was significantly better than that of group B, so the conjecture was verified. To sum up, when using the algorithm to do the style transfer of Yunjin, the color block image can be selected as the content image to obtain better results.

IV. THE DESIGN APPLICATION OF IMITATING YUNJIN STYLE

The generated picture with the modern pattern of Yunjin style obtained by the transfer algorithm is defined as the Imitating Yunjin. The application of Imitating Yunjin style is closely related to that of Yunjin.

In the inheritance of successive dynasties, Nanjing Brocade was mostly dedicated tribute for the royal family, and the selection of color tended to be high purity and high saturation, forming a unique palace style. Wang [6] et al. integrated contemporary design aesthetics with the palace cultural elements of Nanjing Brocade, and completed the innovative design of silk scarf pattern. Song and Wang [7] combined Nanjing Brocade with Olympic dress, which opened up ideas for the dress design of national sports events; Fan and Zhang [8] combbed out the application of traditional patterns and symbols of Nanjing Brocade in cultural creation products, which has practical value for promoting the dissemination of intangible cultural heritage and promoting the development of Nanjing Brocade culture.

In this paper, combined with the application of Nanjing Brocade and the style of Imitating Yunjin, the designs were applied on canvas bag, scented sachet and pillow. The three designs are shown in Figure 6.

In addition to individual textile design, Nanjing Brocade is also suitable for clothing series design. This is because the generated pictures of the same style obtained by the style transfer model are highly similar in style and have a strong sense of series.

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(a) Canvas bag

(b) Scented sachet

(c) Pillow

Figure 6. Textile designs

V. SUMMARY

Our work provides a new method for the digital design of Nanjing Brocade patterns, which can not only protect and inherit Yunjin, but also conform to the modern aesthetic. In addition, through the comparative experiment, it is obvious that taking color block image as content image is more suitable for the transfer of Yunjin. Future work includes optimizing the migration algorithm to get better renderings. What's more, the pattern is only a part of Yunjin culture, further research is essential for the digitization of the craft.

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